SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA5116

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5 **Total Marks:50**

Annexure A

APPLICATION- BASED QUESTIONS

Code of Conduct:

I K.Vineela(192292423) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

K.Vineela
Signature of the student:

Annexure A

APPLICATION- BASED QUESTIONS

1. **Scenario: Event Communication Security**
A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message "MEET AT NOON" is transformed into an encoded version using a fixed shift pattern.
   a. Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.
   b. Explain how an authorized member, aware of the encoding rule, would retrieve the original message.

2. **Scenario: Intercepted Suspicious Message**
A cybersecurity intern intercepts the coded message: "GSRH RH Z HVXIVG" during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.
   a. Deduce the most likely original message.
   b. Justify your answer by analyzing repeating patterns, letter structures, or known cipher characteristics.

3. **Scenario: Secure Messaging System**
A company deploys a keyword-based encryption technique for internal communication. An employee sends the word "HELLO" using the keyword "KEY" to ensure confidentiality.
   a. Demonstrate how the encryption process transforms the message step-by-step.
   b. Explain how the repeating keyword influences each character of the ciphertext.

4. **Scenario: Suspicious Digital Asset Investigation**
During a digital forensic audit, investigators suspect that a company logo file may contain hidden confidential information embedded within it. The file appears normal upon casual inspection.
   a. Analyze how investigators could detect the presence of hidden data within the image.
   b. Recommend two techniques that could strengthen the security of such hidden communications against detection or extraction.

5. **Scenario: Legacy Encryption in Corporate Communication**
A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement. A received message appears as: "WECRLTEERDSOEEFEAOCAIVDEN".
   a. Determine the most probable original message by reversing the encryption process.
   b. Critically evaluate why such a method is vulnerable in modern communication systems.
   c. Propose one practical enhancement to improve the confidentiality of messages in this scenario.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

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CSA5116 - Cryptography & Network Security

1) Scenario : Event communication Security

A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message 'MEET AT NOON' is transformed into an encoded version using a fixed shift pattern.

a). Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.

b). Explain how an authorized member, aware of the encoding rule, would retrieve the original message.

- This scenario uses a caesar cipher,

Plain Text : MEET AT NOON

Key value : 4

a	b	c	d	e	f	g	h	I	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25

Encryption :-

$$c = (p+k) \bmod 26$$

$$c = [12+4] \bmod 26 = 16 \bmod 26$$

$$= 16\ [Q]$$

$$c = [4+4] \bmod 26 = 8 \bmod 26$$

$$= 8\ [I]$$

$C = (4+4) \bmod 26 = 8$ [I]

$C = (19+4) \bmod 26 = 23$ [X]

$C = (0+4) \bmod 26 = 4$ [E]

$C = (19+4) \bmod 26 = 23$ [X]

$C = (13+4) \bmod 26 = 17$ [R]

$C = (14+4) \bmod 26 = 18$ [S]

$C = (14+4) \bmod 26 = 18$ [S]

$C = (13+4) \bmod 26 = 17$ [R]

→ Cipher Text = QIIXRSSR

Decryption :-

$P = (C-K) \bmod 26$

$P = (16-4) \bmod 26 = 12$ [M]

$P = (8-4) \bmod 26 = 4$ [E]

$P = (8+4) \bmod 26 = 4$ [E]

$P = (23-4) \bmod 26 = 19$ [T]

$P = (4-4) \bmod 26 = 0$ [A]

$P = (23-4) \bmod 26 = 19$ [T]

$P = (17-4) \bmod 26 = 13$ [N]

$P = (18-4) \bmod 26 = 14$ [O]

$P = (18-4) \bmod 26 = 14$ [O]

$P = (17-4) \bmod 26 = 14$ [N]

→ plainText = MEET AT NOON

Scenario : Suspicious Digital Asset Investigation.

a) Analyze how investigators could detect the presence of hidden data within the image.

b) Recommended two techniques that could strengthen the security of such hidden communications against detection or extraction.

Detecting hidden data in the image :-

- Visual analysis : zoom and inspect the image for unusual pixels, colors, patterns or distortions.
- File analysis : check the image's file size, metadata, formate and properties. An unusually large file may indicate hidden data.
- Statistical analysis : compare pixel-value patterns with those expected from a normal image. Sometimes it creates detectable statistical anomalies.
- extracting data : If a hiding technique is suspected, investigators can try known steganography extraction methods to determine whether concealed informed is present.

Two techniques to strengthen hidden communication :-

- encryption before hiding

First encrpt the confidental message and then hide the encrypted data inside the image. Even if

investigators extract the hidden data, they cannot easily understand it without the encryption key.

- use a strong steganographic technique.

Instead of simple pixel replacement, use a more sophisticated method that distributes the hidden information carefully and minimizes statistical changes to the image. This makes detection more difficult.

Normal Image → encrypt secret message → Hide encrypted message in image → send image

Image → Detect suspicious changes → extract hidden data → Decrypt → Recover confidential information.